

Balaji Rao

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Education

Stevens Institute of Technology, Hoboken, NJ – ABD

(Expected) December 2026

Ph.D., Systems Engineering

Dissertation Area: Large Language Models (LLMs), LLM Reasoning, Symbolic AI, and Systems Reliability

Advisor: Dr. Carlo Lipizzi

Stevens Institute of Technology, Hoboken, NJ

May 2022

Master of Engineering, Engineering Management

Graduate Certificates: Data Exploration and Visualization for Risk and Decision-Making; Systems Engineering

BNM Institute of Technology, Bengaluru, India

October 2020

Bachelor of Engineering, Electronics and Communication Engineering

Profile

Ph.D. candidate specializing in Systems Engineering, with expertise in Large Language Models (LLMs), Neurosymbolic AI, and scalable machine learning solutions. Skilled in building data-driven solutions and leveraging machine learning methods to address complex challenges. Demonstrated success in fine-tuning SOTA language models, developing novel algorithms, and optimizing NLP workflows, aligning with applied science roles in generative AI.

Skills

Programming Languages: Python, OCaml, Rust, Lean4

Frameworks and Libraries: PyTorch, TensorFlow, scikit-learn, Hugging Face, LangChain, Pinecone, NumPy, Pandas, NLTK, Caffe, OpenCV

Tools and Platforms: Git/GitHub, AWS (Bedrock, SageMaker AI, App Studio), Claude Code, Kiro, Cursor

Machine Learning: Natural Language Processing (NLP), Reinforcement Learning, Neural Theorem Proving (NTP), SMOTE, Knowledge Graphs

Experience

Applied Science Intern - Automated Reasoning (Amazon Web Services)

May 2025 – August 2025

- Built an LLM-assisted HOL Light/s2n-bignum Agentic AI tool for tactic explanations and tactic suggestions, integrating Bedrock, SageMaker, Kendra, and S3. Interfaced via Amazon Q/VS Code Cline through an MCP server.
- Prototyped a neural theorem-proving agent loop (propose→verify→retry on errors); added evaluation/benchmark artifacts and migrated to a locally hosted model to cut token costs and enable rapid iteration.
- Enhanced verification workflows by automating formal correctness proofs and generating natural language descriptions of custom tactics, reducing manual effort for software verification engineers.

Data Science/Data Engineering Intern - Johnson & Johnson

May 2024 - August 2024

- Developed machine learning models to analyze and reduce content fatigue, enhancing healthcare professionals' (HCPs) engagement with promotional emails.
- Implemented a Hidden Markov Model (HMM) for probabilistic predictions of email engagement, utilizing a feature matrix that included temporal data. Integrated use of Gen AI solutions to leverage large language models (LLMs) like Llama-2 to optimize content, improving messaging outcomes.
- Introduced new predictive analytics metrics to provide deeper insights into content fatigue, complementing traditional email engagement metrics.

Research Assistant – Stevens Institute of Technology, Hoboken, NJ

July 2021 – Present

- Co-developed *s2n-bignum-bench*, a HOL Light benchmark of 3,662 proof obligations distilled from the AWS *s2n-bignum* verified cryptography library to evaluate ISA-aware, bit-precise LLM reasoning on low-level code.
- Designed *ProofSeek*, an LLM-based neural theorem proving system that formalizes natural language policies into Isabelle/HOL proofs. Fine-tuned models using a two-stage training regime and evaluated performance using the miniF2F benchmark and real-world cloud security policy datasets (e.g., AWS S3). Deployed a proof assembly module (ProofAug) and applied symbolic verification techniques to extend prior work on formal code validation.
- Developed and evaluated knowledge-augmented LLMs to improve contextual grounding and reduce hallucinations in generative outputs by integrating structured data into prompt pipelines.

- Designed and implemented an automated theorem-proving framework using Isabelle/HOL for verifying policy logic in safety-critical domains. Combined symbolic reasoning with reinforcement learning to improve formal proof generation.
- Proposed and began developing a framework for self-refining LLM systems using reinforcement learning reward tuning to optimize output coherence and factual accuracy across applied generative tasks.
- Built and deployed a domain-specific tutoring assistant (SSE-EduBot) using Llama 2, LangChain, and Pinecone. The system leverages internal academic materials to support personalized student queries across the Systems Engineering curriculum.
- Applied SMOTE to address class imbalance in healthcare datasets and designed machine learning pipelines to support early risk detection for diabetes using electronic health records (EHR).
- Constructed a multimodal classification framework combining EEG and facial expression data to infer user preferences. Evaluated performance across several supervised learning algorithms.

Selected Projects

s2n-bignum-bench: Low-Level Theorem Proving Benchmark for LLMs (HOL Light) December 2025

Packaged 3,662 machine-checkable proof problems derived from the AWS *s2n-bignum* cryptography library into isolated context–query tasks for neural theorem proving evaluation, targeting reasoning. Consists of an end-to-end offline evaluation pipeline that compiles and checks submitted tactics against shipped object-code artifacts, and added integrity defenses with lightweight category taxonomy.

Neural Theorem Proving with LLMs: The ProofSeek Framework March 2025

Developed *ProofSeek*, a neural theorem proving framework combining symbolic reasoning with large language models. The system translates natural language policies into formal logic, generates Isabelle/HOL proofs using a fine-tuned LLM, and constructs verifiable proofs through a heuristic module (ProofAug). Trained using supervised fine-tuning and reinforcement learning with Generalized Reward Propagation Optimization (GRPO), the model was evaluated on the miniF2F dataset and applied to AWS S3 access control policy verification.

LLM Trustworthiness Evaluation with Knowledge Graphs May–September 2024

Developed a methodology to assess the factual reliability of outputs generated by large language models. The approach integrates structured knowledge graphs with human evaluation to identify hallucinated content and improve the transparency of LLM predictions, focusing on domains such as healthcare, defense, and finance.

Formal Verification of Policy Code using Isabelle/HOL April–May 2024

Implemented a verification pipeline that translates LLM-generated policy code into formal logic using the Isabelle/HOL proof assistant and Isar syntax. The project focused on formalizing cloud-based access policies (e.g., AWS S3) and exploring the intersection of symbolic reasoning with language model output.

SSE-EduBot: Department-Wide Tutoring System using LLMs 2023–2025

Designed and deployed an educational chatbot prototype to assist graduate students at Stevens Institute of Technology. Built using Llama 2, LangChain, and Pinecone, the system is aligned with the Systems Engineering curriculum and provides course-aware, domain-specific responses by leveraging instructional content and syllabi.

Validator Node Selection in Proof-of-Stake Blockchain May 2023

Implemented a Game Theory grounded validator selection mechanism based on second-price auctions in a custom PoS blockchain prototype. Developed supporting infrastructure using Go and TCP sockets to enable block proposal broadcasting, peer messaging, and consensus synchronization in a distributed system.

Publications & Conference Papers

- *Steve: LLM Powered ChatBot for Career Progression*, AAAI 2026 – System Demonstration Track
- *SSE-EduBot: Department-Wide Tutoring System Using a Large Language Model*, FIE 2025
- *Neural Theorem Proving: Generating and Structuring Proofs for Formal Verification*, NeSy 2025
- *Anatomy of an AI Economy*, IEEE ISSE 2024
- *Measuring Resilience in Multimodal Transportation Systems*, TRB 2024 (Poster)
- *Identification of Variables Impacting Cascading Failures in Aerospace Systems*, CSER 2024
- *A Game-Theoretic Approach for Validator Selection in PoS Blockchains*, ICoABCD 2023

Fellowships and awards

- *Amazon Post-internship Fellowship* – Financial Support in addition to \$10,000 in AWS credits.
- *Future AI Leaders Program* – The AI Summit New York.